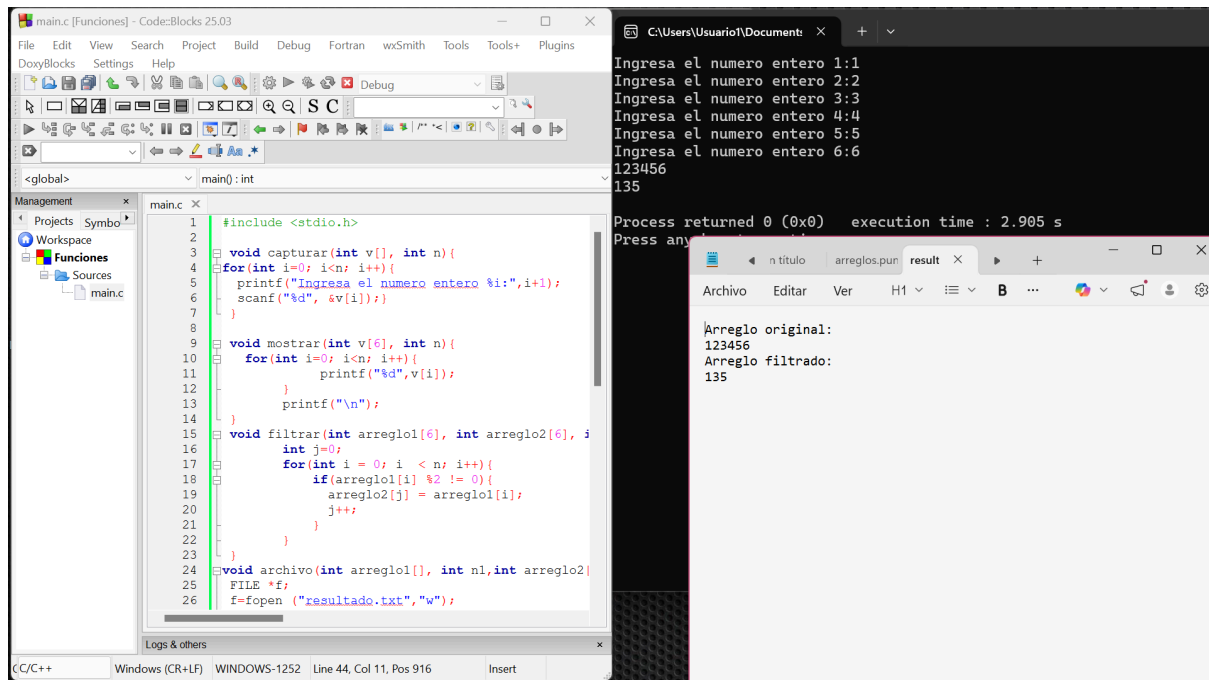
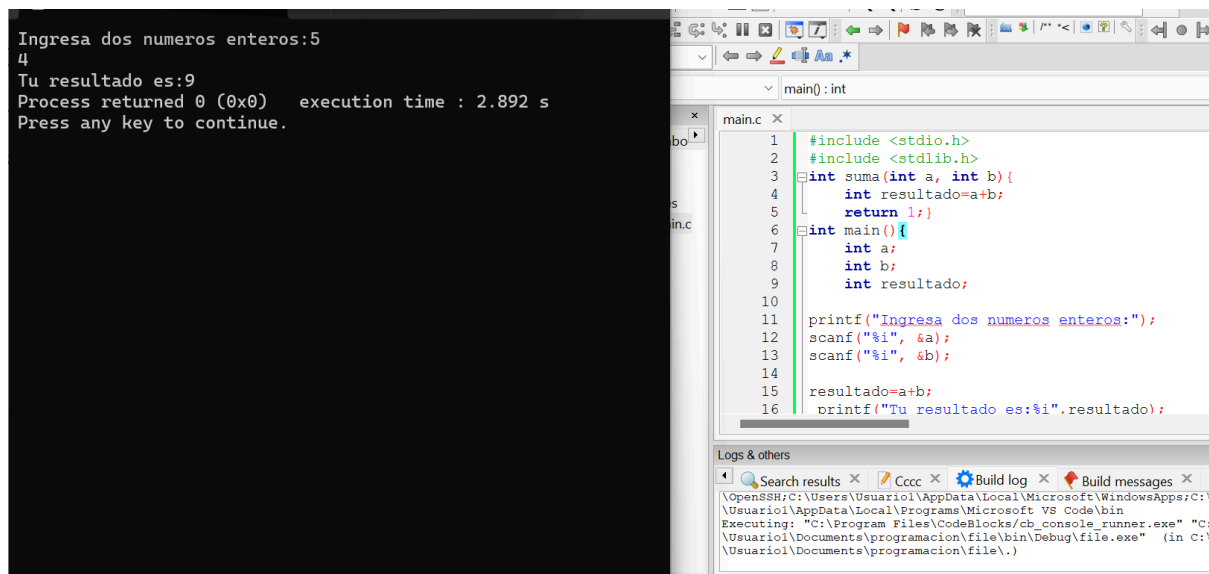


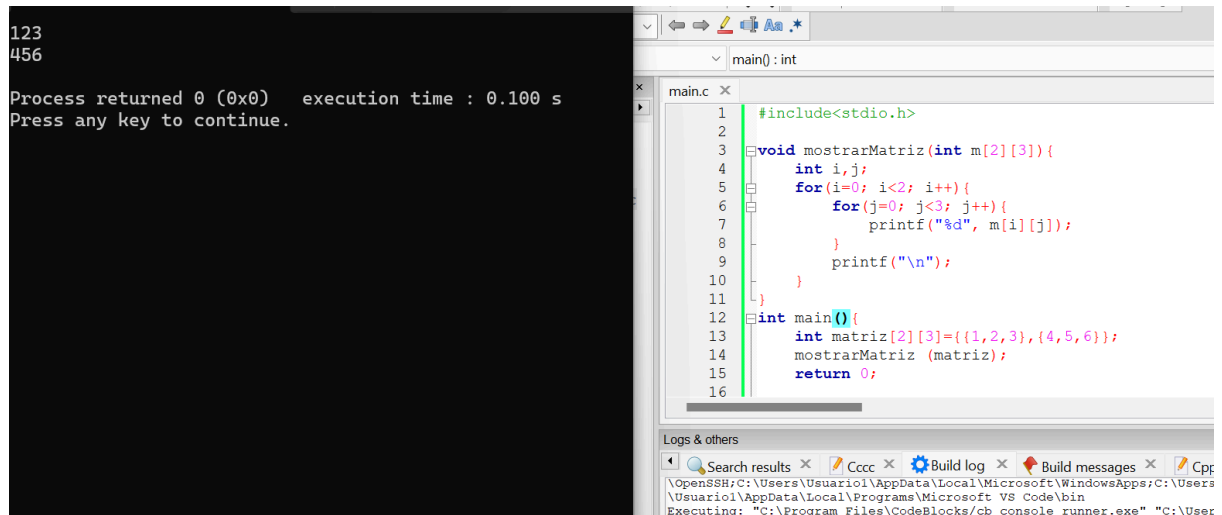
Arreglo



Suma



Mostrar Matriz



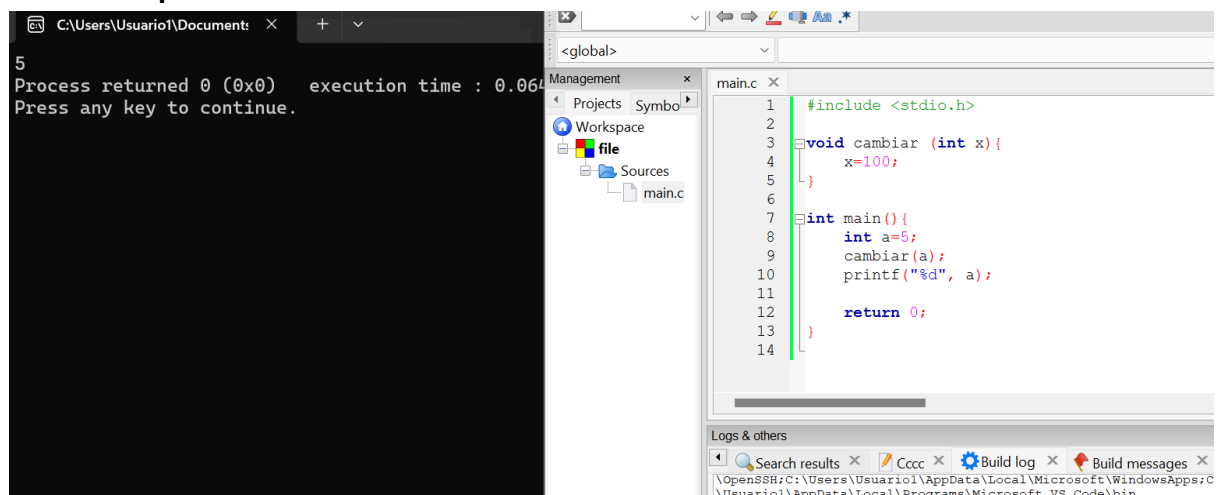
The screenshot shows a C program in Visual Studio Code. The terminal on the left displays the output of the program: a 2x3 matrix with values 1, 2, 3 in the first row and 4, 5, 6 in the second row. The program returns 0 and takes 0.100 seconds to execute. The code in the editor defines a function `mostrarMatriz` that prints a 2x3 matrix and a `main` function that calls it with a matrix of values 1, 2, 3 and 4, 5, 6.

```
123
456

Process returned 0 (0x0)   execution time : 0.100 s
Press any key to continue.
```

```
1 #include<stdio.h>
2
3 void mostrarMatriz(int m[2][3]){
4     int i,j;
5     for(i=0; i<2; i++){
6         for(j=0; j<3; j++){
7             printf("%d", m[i][j]);
8         }
9         printf("\n");
10    }
11 }
12
13 int main(){
14     int matriz[2][3]={1,2,3},{4,5,6};
15     mostrarMatriz (matriz);
16     return 0;
17 }
```

Parametro por valor



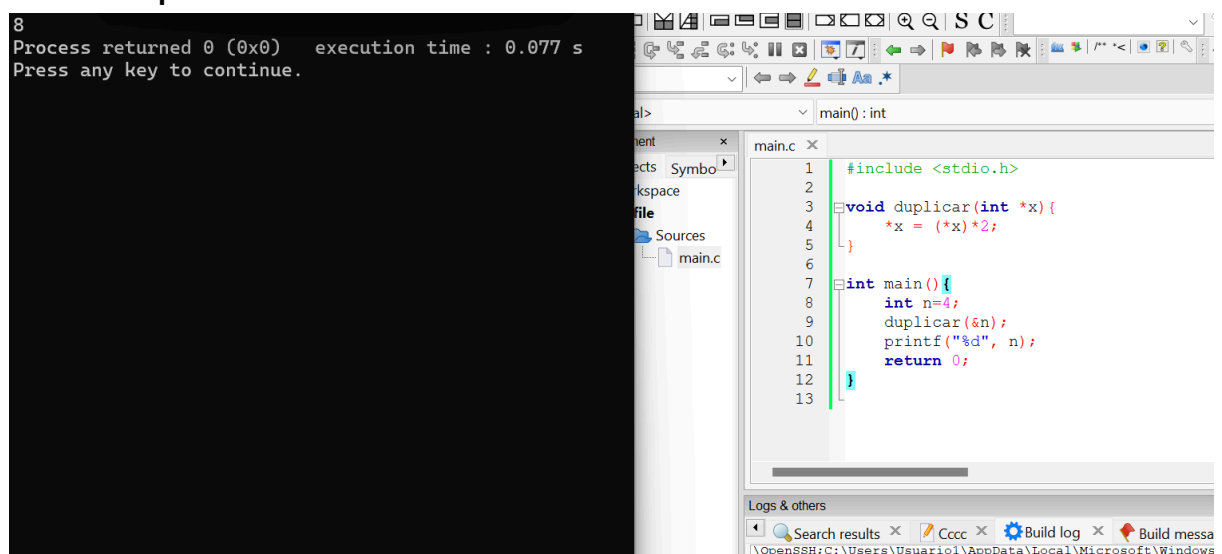
The screenshot shows a C program in Visual Studio Code. The terminal on the left displays the output: the value 5. The program returns 0 and takes 0.064 seconds to execute. The code in the editor defines a function `cambiar` that takes an integer parameter `x` and sets it to 100. The `main` function calls `cambiar` with the value 5 and prints it, showing that the value remains 5 because the parameter is passed by value.

```
5

Process returned 0 (0x0)   execution time : 0.064 s
Press any key to continue.
```

```
1 #include <stdio.h>
2
3 void cambiar (int x){
4     x=100;
5 }
6
7 int main(){
8     int a=5;
9     cambiar(a);
10    printf("%d", a);
11
12    return 0;
13 }
```

Parametro por referencia



The screenshot shows a C program in Visual Studio Code. The terminal on the left displays the output: the value 8. The program returns 0 and takes 0.077 seconds to execute. The code in the editor defines a function `duplicar` that takes a pointer to an integer and doubles the value it points to. The `main` function calls `duplicar` with the address of `n` and prints it, showing that the value of `n` has been doubled from 4 to 8 because the parameter is passed by reference.

```
8

Process returned 0 (0x0)   execution time : 0.077 s
Press any key to continue.
```

```
1 #include <stdio.h>
2
3 void duplicar(int *x){
4     *x = (*x)*2;
5 }
6
7 int main(){
8     int n=4;
9     duplicar(&n);
10    printf("%d", n);
11    return 0;
12 }
13 }
```

Suma Diagonal

The image shows a C++ program running in a console window and a code editor. The console displays the program's execution, including input prompts and the final output of the diagonal sum (8). The code editor shows the source code for the program, which uses a 2D array to store the matrix and calculates the sum of the main diagonal. A file explorer window is also visible, showing the project files.

Console Output:

```

Ingresa los numeros enteros en la fila 1, columna 1: 2
Ingresa los numeros enteros en la fila 1, columna 2: 3
Ingresa los numeros enteros en la fila 1, columna 3: 4
Ingresa los numeros enteros en la fila 2, columna 1: 5
Ingresa los numeros enteros en la fila 2, columna 2: 3
Ingresa los numeros enteros en la fila 2, columna 3: 2
Ingresa los numeros enteros en la fila 3, columna 1: 3
Ingresa los numeros enteros en la fila 3, columna 2: 4
Ingresa los numeros enteros en la fila 3, columna 3: 4
La suma de la diagonal es 8
Process returned 0 (0x0)   execu
Press any key to continue.

```

Code Editor (main.c):

```

43
44     printf("La suma de la diagonal es %i\n", a);
45
46     FILE*f;
47     f=fopen ("tablero.txt","a");
48     fprintf(f,"Tablero:\n");
49     for(int i=0; i<n; i++){
50         for(int j=0; j<n; j++){
51             fprintf(f," %i", arre[i][j]);
52         }
53         fprintf(f,"\n");
54     }
55     printf("La suma de la diagonal es %i\n", a);
56     return 0;

```

File Explorer:

- Resultado.txt
- arreglos.o
- tablero

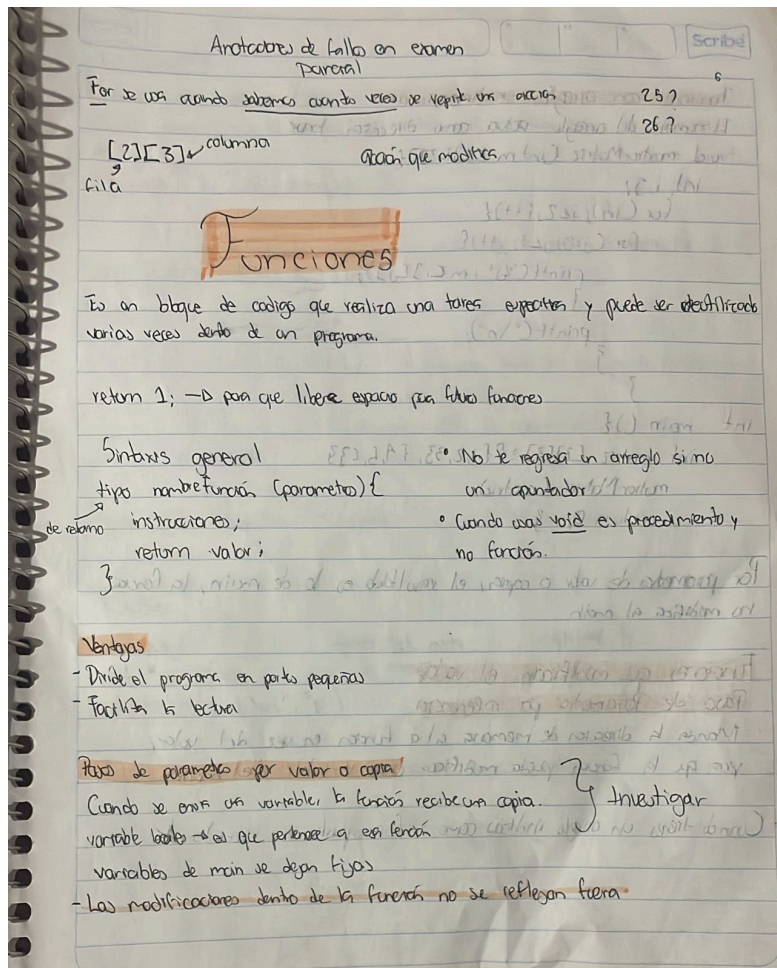
Build Log:

```

Build log
Build messages
CppCheck
E:\Local\Microsoft\WindowsApps\C:\Users\
\Microsoft VS Code\bin
eBlocks\cb_console_runner.exe "C:\Users\
\file\bin\Debug\file.exe" (in C:\Users\
\file\.)

```

NOTAS



Funciones con arreglos unidimensionales

El nombre del arreglo actúa como dirección base

```
void mostrarMatriz (int m [2][3]) {
```

```
    int i, j;
```

```
    for (i=0; i<2; i++) {
```

```
        for (j=0; j<3; j++) {
```

```
            printf("%d", m[i][j]);
```

```
        }
        printf("\n");
    }
```

```
}
```

```
int main () {
```

```
    int matriz [2][3] = { {1, 2, 3}, {4, 5, 6} }
```

```
    mostrarMatriz (matriz);
```

```
    return 0;
```

Por parámetros de valor o copia, el resultado es lo de main, la función no modifica al main

Funciones que modifican el valor

Paso de Parámetro por referencia

Manda la dirección de memoria a la función en vez del valor, por lo que la función pueda modificar una variable del main

Cuando tienes un arreglo indicas como si fueran punteros no necesitas &